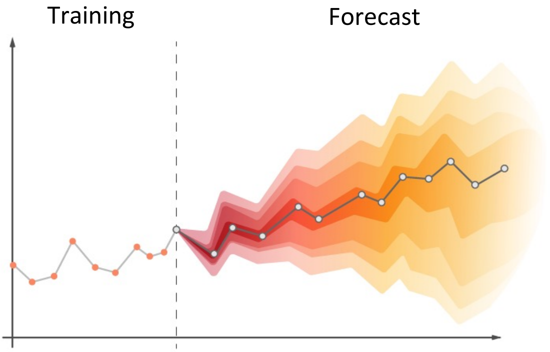
Final project

Time-series data and application to stock markets

Introduction

Time series data is basically a collection of measurements recorded over time.
Examples:

- Temperature data points taken every hour.
- Stock prices taken every day.



Training and forecasting with time-series data.

Introduction

Time-series data has patterns.

Question: What if you can find the patterns hidden in the stock market?

Project overview

The project aims at getting students familiar with time-series data and its applications by analyzing and deriving practical solutions using predictive analytics for stock markets.

There are two types of stock markets considered in this project:

- International stock market (Nasdaq).
- Vietnam stock market (HOSE, HNX and UPCOM).

Nasdaq dataset

Data features of Nasdaq dataset:

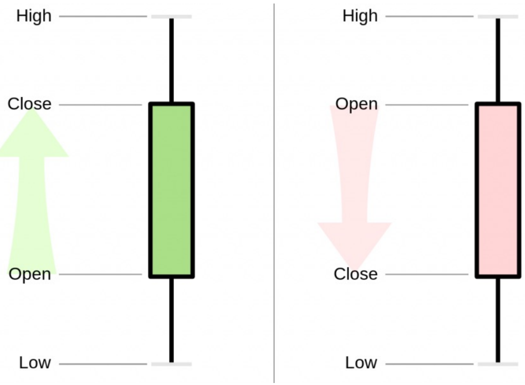
- **Date**: data collection date.
- **Low**: refers to the lowest price of the period.
- **Open**: refers to the price at which a stock started trading when the opening bell rang.
- **Volume**: refers to the total number of shares traded in a day.
- **High**: refers to the highest price at which a stock is traded during a period.
- **Close**: refers to cost of shares at the end of the day.
- **Adjusted close**: considers other factors like dividends, stock splits, and new stock offerings. Since the adjusted closing price begins where the closing price ends, it can be called a more accurate measure of stocks' value.

As of Dec. 12, 2022, the Nasdaq dataset contains 1,564 companies and takes 841 MBs of csv text data after extraction.

Nasdaq dataset

Increasing price

Decreasing price



Nasdaq dataset

Example of stock price data of APPLE company (APPL) in the Nasdaq dataset.

AAPL

Date	Low	Open	Volume	High	Close	Adjusted Close
12-12-1980	0.1283479928970337	0.1283479928970337	469033600	0.1289059966802597	0.1283479928970337	0.09987381845712662
15-12-1980	0.12165199965238571	0.12221000343561172	175884800	0.12221000343561172	0.12165199965238571	0.09466329962015152
16-12-1980	0.11272300034761429	0.1132809966802597	105728000	0.1132809966802597	0.11272300034761429	0.08771524578332901
17-12-1980	0.11551299691200256	0.11551299691200256	86441600	0.11607100069522858	0.11551299691200256	0.08988628536462784
18-12-1980	0.11886200308799744	0.11886200308799744	73449600	0.11941999942064285	0.11886200308799744	0.09249227494001389
19-12-1980	0.12611599266529083	0.12611599266529083	48630400	0.12667399644851685	0.12611599266529083	0.09813698381185532
22-12-1980	0.1322540044784546	0.1322540044784546	37363200	0.1328130066394806	0.1322540044784546	0.10291323065757751
23-12-1980	0.13783499598503113	0.13783499598503113	46950400	0.13839299976825714	0.13783499598503113	0.10725609958171844
24-12-1980	0.14508900046348572	0.14508900046348572	48003200	0.14564700424671173	0.14508900046348572	0.11290080845355988
26-12-1980	0.15848200023174286	0.15848200023174286	55574400	0.15904000401496887	0.15848200023174286	0.1233224868774414
29-12-1980	0.16071400046348572	0.16071400046348572	93161600	0.16127200424671173	0.16071400046348572	0.1250593215227127
30-12-1980	0.156808003783226	0.15736599266529083	68880000	0.15736599266529083	0.156808003783226	0.1220199316740036
31-12-1980	0.1523440033197403	0.1529020071029663	35750400	0.1529020071029663	0.1523440033197403	0.11854627728462219
02-01-1981	0.15401799976825714	0.15401799976825714	21660800	0.15513400733470917	0.15401799976825714	0.11984889954328537
05-01-1981	0.15067000687122345	0.15122799575328827	35728000	0.15122799575328827	0.15067000687122345	0.11724361032247543
06-01-1981	0.1439729928970337	0.1445309966802597	45158400	0.1445309966802597	0.1439729928970337	0.11203235387802124
07-01-1981	0.13783499598503113	0.13839299976825714	55686400	0.13839299976825714	0.13783499598503113	0.10725609958171844

Vietnam stock market dataset

Data features:

- **TradingDate**: data collection date.
- **Low**: refers to the lowest price of the period.
- **Open**: refers to the price at which a stock started trading when the opening bell rang.
- **Volume**: refers to the total number of shares traded in a day.
- **High**: refers to the highest price at which a stock is traded during a period.
- **Close**: refers to cost of shares at the end of the day.

The dataset contains three Vietnam stock exchanges:

- HoSE (VNIndex).
- HNX (HNXIndex).
- Upcom (UpcomIndex).

Vietnam stock market dataset

Example of stock price data of Hoà Phát Group (HPG) in the Nasdaq dataset.

HPG-VNINDEX-History

	Open	High	Low	Close	Volume	TradingDate
0	2663.0	2704.0	2267.0	2642.0	1306330	2007-11-15
1	2517.0	2517.0	2517.0	2517.0	248510	2007-11-16
2	2392.0	2392.0	2392.0	2392.0	120480	2007-11-19
3	2288.0	2288.0	2288.0	2288.0	58710	2007-11-20
4	2184.0	2371.0	2184.0	2371.0	728080	2007-11-21
5	2288.0	2371.0	2288.0	2371.0	266040	2007-11-22
6	2371.0	2371.0	2288.0	2288.0	106370	2007-11-23

Vietnam stock market dataset

List of features

	Open	High	Low	Close	Volume	TradingDate
0	2663.0	2704.0	2267.0	2642.0	1306330	2007-11-15
1	2517.0	2517.0	2517.0	2517.0	248510	2007-11-16
2	2392.0	2392.0	2392.0	2392.0	120480	2007-11-19
3	2288.0	2288.0	2288.0	2288.0	58710	2007-11-20
4	2184.0	2371.0	2184.0	2371.0	728080	2007-11-21
5	2288.0	2371.0	2288.0	2371.0	266040	2007-11-22
6	2371.0	2371.0	2288.0	2288.0	106370	2007-11-23

Vietnam stock market dataset

Vietnam stock exchanges also have other information data available:

- dividend-history: historical dividends of companies.
- financial-ratio: financial health of companies.
- industry-analysis: information about companies of the same industry.

HPG-VNINDEX-Dividend

	exerciseDate	cashYear	cashDividendPercentage	issueMethod
0	17/06/22	2021	0.05	cash
1	17/06/22	2022	0.3	share
2	31/05/21	2021	0.35	share
3	31/05/21	2020	0.05	cash
4	29/07/20	2019	0.05	cash
5	29/07/20	2020	0.2	share
6	04/06/19	2019	0.3	share

	ticker	marcap	price	numberOfDays	priceToEarning	peg	priceToBook	valueBeforeEbitda	dividend	roe	roa	ebitOnInterest
0	HPG	121529	20900	1	14.3	-0.2	1.3	8.8	0.0	91	49	-2.1
1	HSG	9300	15550	2	-8.7	0.1	0.9	35.2	0.0	-0.1	-55	-12.7
2	NKG	3975	15100	1	-59.6	0.6	0.7	16.3	0.0	-12	-5	-4.1
3	TVN	3648	5380	1	-4.4	0.0	0.4	54.6	0.0	-91	-33	0.1
4	SHI	2266	14000	1	38.9	-2.2	1.3	14.5	0.0	41	9	1.1
5	DTL	1667	27500	0	-13.8	0.0	1.8	21.2	0.0	-121	-57	-2.2
6	POM	1448	5200	0	-1.2	0.0	0.6	-11.9	0.0	-385	-101	-1.9
7	TIS	874	4753	0	-90.8	0.8	0.5	20.1	0.0	-5	-1	0.5
8	SMC	762	10350	1	-1.3	0.0	0.4	-16.2	0.0	-283	-66	-6.0
9	TLH	729	7140	1	134.2	-1.4	0.4	12.1	0.0	3	1	-1.9
10	VGS	615	12700	1	6.2	-0.3	0.7	8.4	0.0	114	47	1.9
11	CBI	392	9105	1	50.5	-0.5	0.9	6.1	0.0	18	4	0.4
12	TTS	361	7100	0	132.7	-1.9	1.2	7.9	0.0	9	2	0.7
13	TNA	298	6050	-3	13.4	3.0	0.5	12.3	0.0	38	9	0.3
14	TNI	151	2880	1	48.4	-0.4	0.3	8.4	0.0	6	3	-11.5
15	SHA	135	4050	1	7.5	5.7	0.3	8.9	0.0	45	18	1.1

Industry analysis

	ticker	quarter	year	priceToEarning	priceToBook	valueBeforeEbitda	dividend	roe	roa	daysReceivable	daysInventory	daysPayable	ebitOnInterest
0	HPG	4	2022	14.3	1.3	8.8		91	49	50	112	54.0	-2.1
1	HPG	3	2022	6.9	1.3	6.0		197	0.1	55	123	52.0	0.1
2	HPG	2	2022	4.3	1.3	3.6		346	164	44	140	65.0	7.9
3	HPG	1	2022	5.6	2.0	3.6		433	0.22	45	104	42.0	15.5
4	HPG	4	2021	5.9	2.3	5.5		461	223	31	115	51.0	13.3
5	HPG	3	2021	7.3	2.8	5.8		456	217	44	132	49.0	16.4
6	HPG	2	2021	9.0	3.1	8.7		398	184	29	129	56.0	17.8
7	HPG	1	2021	8.4	2.3	9.5		313	147	30	110	34.0	12.2
8	HPG	4	2020	10.0	2.3	10.1		252	115	17	117	40.0	8.7
9	HPG	3	2020	8.0	1.6	9.4		213	0.1	30	109	35.0	8.7
10	HPG	2	2020	8.2	1.4	9.1		0.18	85	31	111	41.0	6.3
11	HPG	1	2020	5.6	0.9	8.1		174	83	29	106	40.0	7.0
12	HPG	4	2019	8.4	1.3	7.7		0.17	83	14	117	52.0	9.1
13	HPG	3	2019	8.0	1.3	8.8		174	88	16	125	35.0	8.7
14	HPG	2	2019	7.9	1.4	7.1		199	103	14	108	37.0	12.3
15	HPG	1	2019	8.2	1.6	6.9		214	117	14	94	42.0	12.5

Financial ratio

Project evaluation

Task	Subtask	Description	Grade	
Task 1 - Nasdaq stock price prediction (Nasdaq dataset)	Task 1.1	Nasdaq multi-feature extension	5%	15%
	Task 1.2	Nasdaq k^{th} day forecast	5%	
	Task 1.3	Nasdaq k days forecast	5%	
Task 2 - Vietnam stock price prediction (Vietnam dataset)	Task 2.1	Vietnam multi-feature extension	5%	15%
	Task 2.2	Vietnam k^{th} day forecast	5%	
	Task 2.3	Vietnam k days forecast	5%	
Task 3 - Trading signal identification for Vietnam market	Task 3.1	Buying signal identification	10%	20%
	Task 3.2	Selling signal identification	10%	
Task 4 - Portfolio composition, risk management and portfolio optimization for Vietnam market	Task 4.1	Portfolio composition	10%	30%
	Task 4.2	Risk management	10%	
	Task 4.3	Portfolio optimization	10%	
Task 5 (Extra credit) - Industry standard for deployment and ease of use	Task 5.1	Model deployment	5%	30%
	Task 5.2	Model as SaaS	10%	
	Task 5.3	AI engineering workflow	15%	
Task 6 - Report			20%	20%
Total			130%	

Project instructions

Project instructions

Disclaimer: These instructions are merely suggestions for solving tasks. As the final project is *free-form*, you are encouraged to tackle the tasks in your own way.

Task 1.1 - Nasdaq multi-feature extension

One-feature input

Output label

Multi-feature input

Modify the demo code so that the Nasdaq stock price prediction model utilizes **multiple features**, such as Low, High, Open, Close, Adjusted Close prices, and Volume, rather than relying solely on one feature (Open price).

AAPL

Date	Low	Open	Volume	High	Close	Adjusted Close
12-12-1980	0.1283479928970337	0.1283479928970337	469033600	0.1289059966802597	0.1283479928970337	0.09987381845712662
15-12-1980	0.12165199965238571	0.12221000343561172	175884800	0.12221000343561172	0.12165199965238571	0.09466329962015152
16-12-1980	0.11272300034761429	0.1132809966802597	105728000	0.1132809966802597	0.11272300034761429	0.08771524578332901
17-12-1980	0.11551299691200256	0.11551299691200256	86441600	0.11607100069522858	0.11551299691200256	0.08988628536462784
18-12-1980	0.11886200308799744	0.11886200308799744	73449600	0.11941999942064285	0.11886200308799744	0.09249227494001389
19-12-1980	0.12611599266529083	0.12611599266529083	48630400	0.12667399644851685	0.12611599266529083	0.09813698381185532
22-12-1980	0.1322540044784546	0.1322540044784546	37363200	0.1328130066394806	0.1322540044784546	0.10291323065757751
23-12-1980	0.13783499598503113	0.13783499598503113	46950400	0.13839299976825714	0.13783499598503113	0.10725609958171844
24-12-1980	0.14508900046348572	0.14508900046348572	48003200	0.14564700424671173	0.14508900046348572	0.11290080845355988
26-12-1980	0.15848200023174286	0.15848200023174286	55574400	0.15904000401496887	0.15848200023174286	0.1233224868774414
29-12-1980	0.16071400046348572	0.16071400046348572	93161600	0.16127200424671173	0.16071400046348572	0.1250593215227127
30-12-1980	0.156808003783226	0.15736599266529083	68880000	0.15736599266529083	0.156808003783226	0.1220199316740036
31-12-1980	0.1523440033197403	0.1529020071029663	35750400	0.1529020071029663	0.1523440033197403	0.11854627728462219
02-01-1981	0.15401799976825714	0.15401799976825714	21660800	0.15513400733470917	0.15401799976825714	0.11984889954328537
05-01-1981	0.15067000687122345	0.15122799575328827	35728000	0.15122799575328827	0.15067000687122345	0.11724361032247543
06-01-1981	0.1439729928970337	0.1445309966802597	45158400	0.1445309966802597	0.1439729928970337	0.11203235387802124
07-01-1981	0.13783499598503113	0.13839299976825714	55686400	0.13839299976825714	0.13783499598503113	0.10725609958171844

Task 1.2 - Nasdaq k^{th} day forecast

Update the demo code to enable the Nasdaq stock price prediction model to forecast the price on the k^{th} day in the future, such as the 3rd day or the 7th day ahead, instead of just the next day.

Input features

3rd day forecast

7th day forecast

AAPL

Date	Low	Open	Volume	High	Close	Adjusted Close
12-12-1980	0.1283479928970337	0.1283479928970337	469033600	0.1289059966802597	0.1283479928970337	0.09987381845712662
15-12-1980	0.12165199965238571	0.12221000343561172	175884800	0.12221000343561172	0.12165199965238571	0.09466329962015152
16-12-1980	0.11272300034761429	0.1132809966802597	105728000	0.1132809966802597	0.11272300034761429	0.08771524578332901
17-12-1980	0.11551299691200256	0.11551299691200256	86441600	0.11607100069522858	0.11551299691200256	0.08988628536462784
18-12-1980	0.11886200308799744	0.11886200308799744	73449600	0.11941999942064285	0.11886200308799744	0.09249227494001389
19-12-1980	0.12611599266529083	0.12611599266529083	48630400	0.12667399644851685	0.12611599266529083	0.09813698381185532
22-12-1980	0.1322540044784546	0.1322540044784546	37363200	0.1328130066394806	0.1322540044784546	0.10291323065757751
23-12-1980	0.13783499598503113	0.13783499598503113	46950400	0.13839299976825714	0.13783499598503113	0.10725609958171844
24-12-1980	0.14508900046348572	0.14508900046348572	48003200	0.14564700424671173	0.14508900046348572	0.11290080845355988
26-12-1980	0.15848200023174286	0.15848200023174286	55574400	0.15904000401496887	0.15848200023174286	0.1233224868774414
29-12-1980	0.16071400046348572	0.16071400046348572	93161600	0.16127200424671173	0.16071400046348572	0.1250593215227127
30-12-1980	0.156808003783226	0.15736599266529083	68880000	0.15736599266529083	0.156808003783226	0.1220199316740036
31-12-1980	0.1523440033197403	0.1529020071029663	35750400	0.1529020071029663	0.1523440033197403	0.11854627728462219
02-01-1981	0.15401799976825714	0.15401799976825714	21660800	0.15513400733470917	0.15401799976825714	0.11984889954328537
05-01-1981	0.15067000687122345	0.15122799575328827	35728000	0.15122799575328827	0.15067000687122345	0.11724361032247543
06-01-1981	0.1439729928970337	0.1445309966802597	45158400	0.1445309966802597	0.1439729928970337	0.11203235387802124
07-01-1981	0.13783499598503113	0.13839299976825714	55686400	0.13839299976825714	0.13783499598503113	0.10725609958171844

Task 1.3 – Nasdaq k days forecast

Extend the demo code so that the Nasdaq stock price prediction model is capable of predicting k consecutive days ahead.

Input features

3 days forecast

7 days forecast

AAPL

Date	Low	Open	Volume	High	Close	Adjusted Close
12-12-1980	0.1283479928970337	0.1283479928970337	469033600	0.1289059966802597	0.1283479928970337	0.09987381845712662
15-12-1980	0.12165199965238571	0.12221000343561172	175884800	0.12221000343561172	0.12165199965238571	0.09466329962015152
16-12-1980	0.11272300034761429	0.1132809966802597	105728000	0.1132809966802597	0.11272300034761429	0.08771524578332901
17-12-1980	0.11551299691200256	0.11551299691200256	86441600	0.11607100069522858	0.11551299691200256	0.08988628536462784
18-12-1980	0.11886200308799744	0.11886200308799744	73449600	0.11941999942064285	0.11886200308799744	0.09249227494001389
19-12-1980	0.12611599266529083	0.12611599266529083	48630400	0.12667399644851685	0.12611599266529083	0.09813698381185532
22-12-1980	0.1322540044784546	0.1322540044784546	37363200	0.1328130066394806	0.1322540044784546	0.10291323065757751
23-12-1980	0.13783499598503113	0.13783499598503113	46950400	0.13839299976825714	0.13783499598503113	0.10725609958171844
24-12-1980	0.14508900046348572	0.14508900046348572	48003200	0.14564700424671173	0.14508900046348572	0.11290080845355988
26-12-1980	0.15848200023174286	0.15848200023174286	55574400	0.15904000401496887	0.15848200023174286	0.1233224868774414
29-12-1980	0.16071400046348572	0.16071400046348572	93161600	0.16127200424671173	0.16071400046348572	0.1250593215227127
30-12-1980	0.156808003783226	0.15736599266529083	68880000	0.15736599266529083	0.156808003783226	0.1220199316740036
31-12-1980	0.1523440033197403	0.1529020071029663	35750400	0.1529020071029663	0.1523440033197403	0.11854627728462219
02-01-1981	0.15401799976825714	0.15401799976825714	21660800	0.15513400733470917	0.15401799976825714	0.11984889954328537
05-01-1981	0.15067000687122345	0.15122799575328827	35728000	0.15122799575328827	0.15067000687122345	0.11724361032247543
06-01-1981	0.1439729928970337	0.1445309966802597	45158400	0.1445309966802597	0.1439729928970337	0.11203235387802124
07-01-1981	0.13783499598503113	0.13839299976825714	55686400	0.13839299976825714	0.13783499598503113	0.10725609958171844

Task 2.1 - Vietnam multi-feature extension

Modify the demo code so that the Vietnam stock price prediction model utilizes **multiple features**, such as Low, High, Open, Close prices, and Volume, rather than relying solely on one feature (Open price).

One-feature input

Output label

Multi-feature input

HPG-VNINDEX-History

	Open	High	Low	Close	Volume	TradingDate
0	2663.0	2704.0	2267.0	2642.0	1306330	2007-11-15
1	2517.0	2517.0	2517.0	2517.0	248510	2007-11-16
2	2392.0	2392.0	2392.0	2392.0	120480	2007-11-19
3	2288.0	2288.0	2288.0	2288.0	58710	2007-11-20
4	2184.0	2371.0	2184.0	2371.0	728080	2007-11-21
5	2288.0	2371.0	2288.0	2371.0	266040	2007-11-22
6	2371.0	2371.0	2288.0	2288.0	106370	2007-11-23
7	2247.0	2288.0	2226.0	2267.0	135430	2007-11-26
8	2247.0	2247.0	2184.0	2184.0	262120	2007-11-27
9	2205.0	2226.0	2205.0	2226.0	131020	2007-11-28
10	2226.0	2247.0	2205.0	2226.0	128110	2007-11-29
11	2205.0	2205.0	2184.0	2205.0	114720	2007-11-30
12	2226.0	2226.0	2205.0	2205.0	68710	2007-12-03
13	2184.0	2205.0	2184.0	2184.0	125040	2007-12-04
14	2143.0	2184.0	2143.0	2163.0	112160	2007-12-05
15	2205.0	2226.0	2163.0	2184.0	76580	2007-12-06
16	2143.0	2184.0	2122.0	2122.0	133400	2007-12-07
17	2080.0	2101.0	2028.0	2028.0	156420	2007-12-10

Task 2.2 - Vietnam k^{th} day forecast

Update the demo code to enable the Vietnam stock price prediction model to forecast the price on the k^{th} day in the future, such as the 3rd day or the 7th day ahead, instead of just the next day.

Input features

3rd day forecast

7th day forecast

HPG-VNINDEX-History

	Open	High	Low	Close	Volume	TradingDate
0	2663.0	2704.0	2267.0	2642.0	1306330	2007-11-15
1	2517.0	2517.0	2517.0	2517.0	248510	2007-11-16
2	2392.0	2392.0	2392.0	2392.0	120480	2007-11-19
3	2288.0	2288.0	2288.0	2288.0	58710	2007-11-20
4	2184.0	2371.0	2184.0	2371.0	728080	2007-11-21
5	2288.0	2371.0	2288.0	2371.0	266040	2007-11-22
6	2371.0	2371.0	2288.0	2288.0	106370	2007-11-23
7	2247.0	2288.0	2226.0	2267.0	135430	2007-11-26
8	2247.0	2247.0	2184.0	2184.0	262120	2007-11-27
9	2205.0	2226.0	2205.0	2226.0	131020	2007-11-28
10	2226.0	2247.0	2205.0	2226.0	128110	2007-11-29
11	2205.0	2205.0	2184.0	2205.0	114720	2007-11-30
12	2226.0	2226.0	2205.0	2205.0	68710	2007-12-03
13	2184.0	2205.0	2184.0	2184.0	125040	2007-12-04
14	2143.0	2184.0	2143.0	2163.0	112160	2007-12-05
15	2205.0	2226.0	2163.0	2184.0	76580	2007-12-06
16	2143.0	2184.0	2122.0	2122.0	133400	2007-12-07
17	2080.0	2101.0	2028.0	2028.0	156420	2007-12-10

Task 2.3 - Vietnam k days forecast

Extend the demo code so that the Vietnam stock price prediction model is capable of predicting k consecutive days ahead.

Input features

3 days forecast

7 days forecast

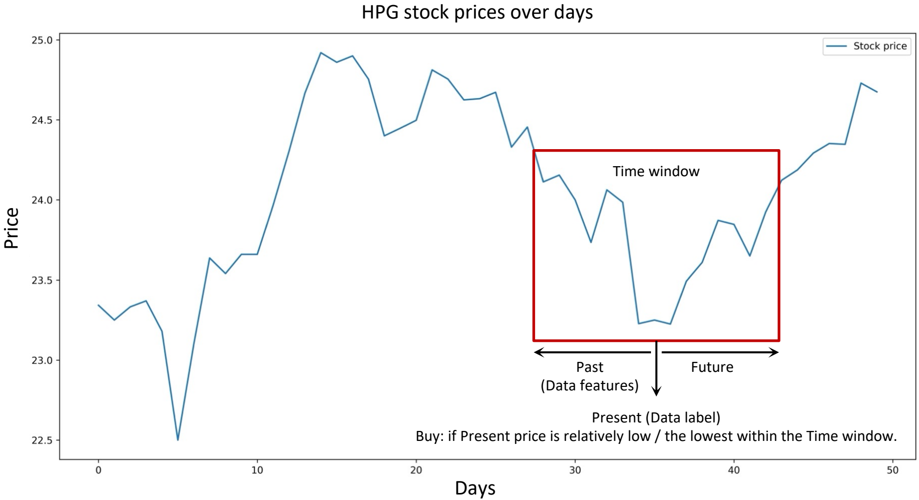
HPG-VNINDEX-History

	Open	High	Low	Close	Volume	TradingDate
0	2663.0	2704.0	2267.0	2642.0	1306330	2007-11-15
1	2517.0	2517.0	2517.0	2517.0	248510	2007-11-16
2	2392.0	2392.0	2392.0	2392.0	120480	2007-11-19
3	2288.0	2288.0	2288.0	2288.0	58710	2007-11-20
4	2184.0	2371.0	2184.0	2371.0	728080	2007-11-21
5	2288.0	2371.0	2288.0	2371.0	266040	2007-11-22
6	2371.0	2371.0	2288.0	2288.0	106370	2007-11-23
7	2247.0	2288.0	2226.0	2267.0	135430	2007-11-26
8	2247.0	2247.0	2184.0	2184.0	262120	2007-11-27
9	2205.0	2226.0	2205.0	2226.0	131020	2007-11-28
10	2226.0	2247.0	2205.0	2226.0	128110	2007-11-29
11	2205.0	2205.0	2184.0	2205.0	114720	2007-11-30
12	2226.0	2226.0	2205.0	2205.0	68710	2007-12-03
13	2184.0	2205.0	2184.0	2184.0	125040	2007-12-04
14	2143.0	2184.0	2143.0	2163.0	112160	2007-12-05
15	2205.0	2226.0	2163.0	2184.0	76580	2007-12-06
16	2143.0	2184.0	2122.0	2122.0	133400	2007-12-07
17	2080.0	2101.0	2028.0	2028.0	156420	2007-12-10

Task 3.1 - Buying signal identification

Build a model to identify potential entry points for *buying* stocks, i.e., the model outputs an indication that can be a score or a probability suggesting it might be a good time to buy. Justify the way you build the model.

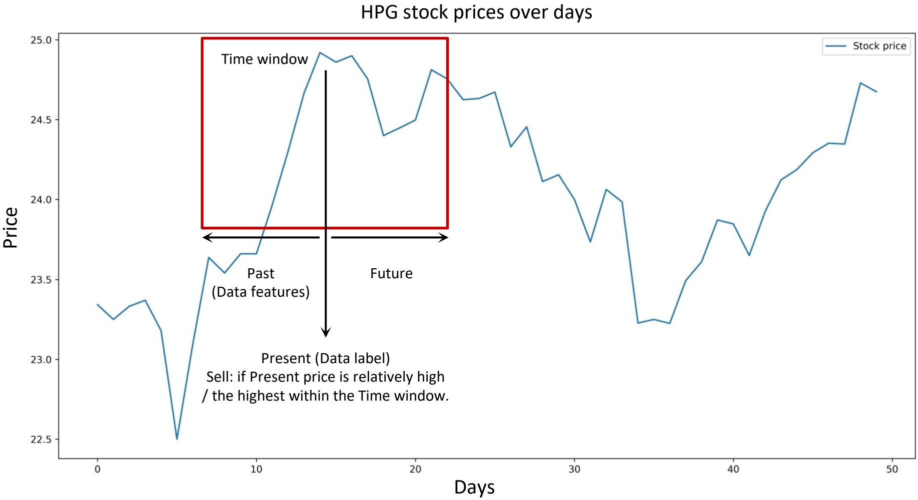
Time window



Task 3.2 - Selling signal identification

Time window

Build a model to identify potential entry points for selling stocks, i.e., the model outputs an indication that can be a score or a probability suggesting it might be a good time to sell. Justify the way you build the model.



Task 4.1 - Profitable stock selection

Select a list of profitable Vietnamese companies to include in the portfolio. Evaluate the projected profit potential of each chosen company within a designated time frame. Explain the methodology used for portfolio optimization and profit estimation.

Price increase

Price decrease

Companies	Price in 7 days
VNM	+22%
MBB	-6%
VHM	+9%
VCB	-1%
VRE	+12%
NLG	-5%
FPT	+16%

Top profitable companies

Task 4.2 - Risk management

Identify risky companies that should be excluded from the portfolio. Explain the risk scoring methodology in your risk scoring model.

Price increase

Price decrease

Companies	Price in 7 days
VNM	+22%
MBB	-6%
VHM	+9%
VCB	-1%
VRE	+12%
NLG	-5%
FPT	+16%

Top risky companies

Task 4.3 - Portfolio composition

Outline a method to determine the optimal investment allocation within the portfolio, i.e., maximize expected return while minimizing risk. Justify your chosen approach for portfolio construction.

Price increases

Price decreases

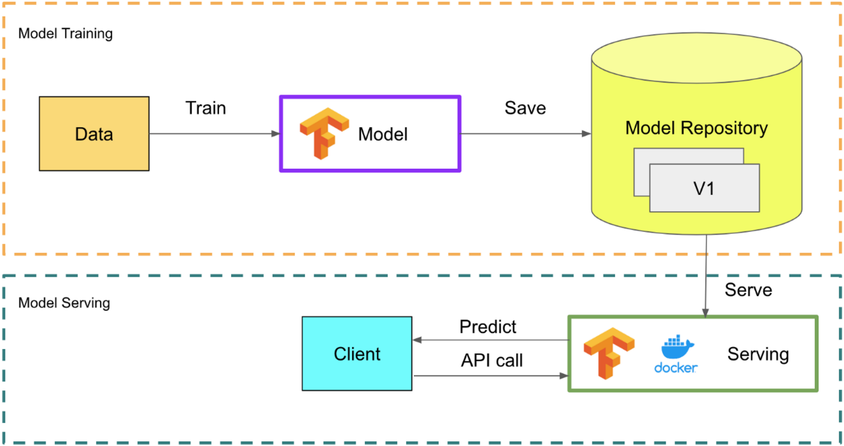
Companies
VNM
MBB
VHM
VCB
VRE
NLG
FPT

Questions:

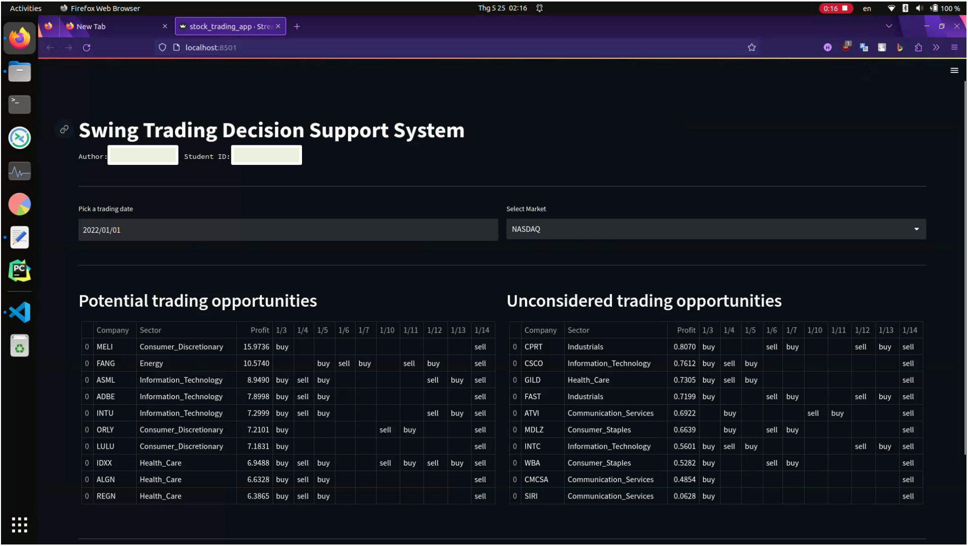
- What other factors should contribute to the profit of a company? Hint: *dividend*.
- What other factors should contribute to the risk mitigation? Hint: *industry sector*.

Task 5.1 (Extra credit) - Model deployment

Deploy the prediction models as API services. Some keywords to research are Tensorflow Serving (TFServing), REST APIs, gRPC.



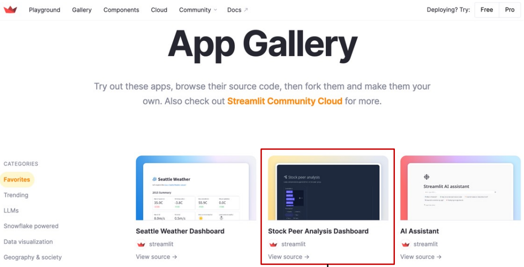
Task 5.2 (Extra credit) - Model as SaaS



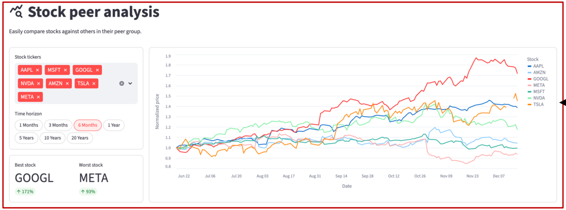
Stock price prediction as SaaS built by a FUV student

Task 5.2 (Extra credit) - Model as SaaS

To deploy a web-based interface, you may consider Python frameworks that are designed for quickly building interactive applications for data science and machine learning models, such as Streamlit or Gradio. **Streamlit** (https://streamlit.io/) is recommended due to its ease of use and developer friendly design. However, the choice of framework is ultimately up to you, depending on which tool you prefer to explore.



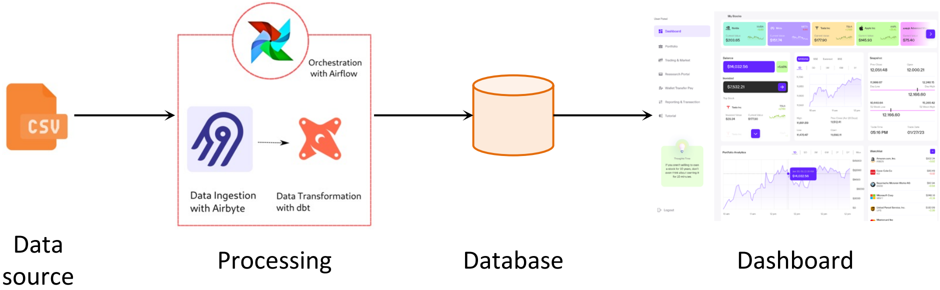
Streamlit App Gallery



Interface of Steamlit Stock Analysis web app

Task 5.3 (Extra credit) - AI automation workflow

Design an engineering flow to automate the tasks. Some keywords to research are SQL, MongoDB, Airflow, Airbyte, dbt.

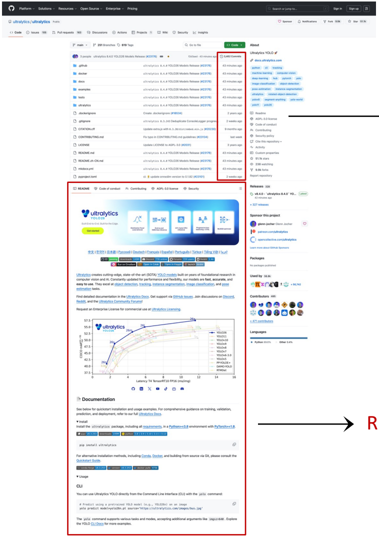


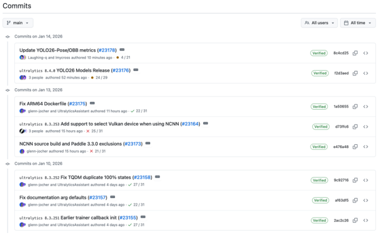
Task 6.1 – Report Writing

Report should describe the journey about your experiments, observations, findings and conclusions from Task 1 to Task 5; it must be at least 2,500 words in length.

Task 6.2 & Task 6.3 – GitHub Repository & README

GitHub Repository contains organized, reproducible code for Tasks 1–5, meaningful commit history, and a clear README with setup and execution instructions.



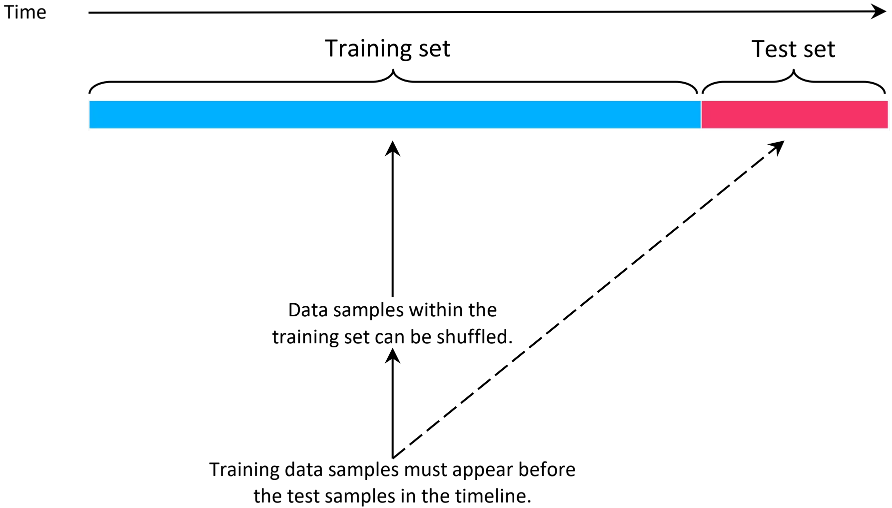


Git Commit History

README

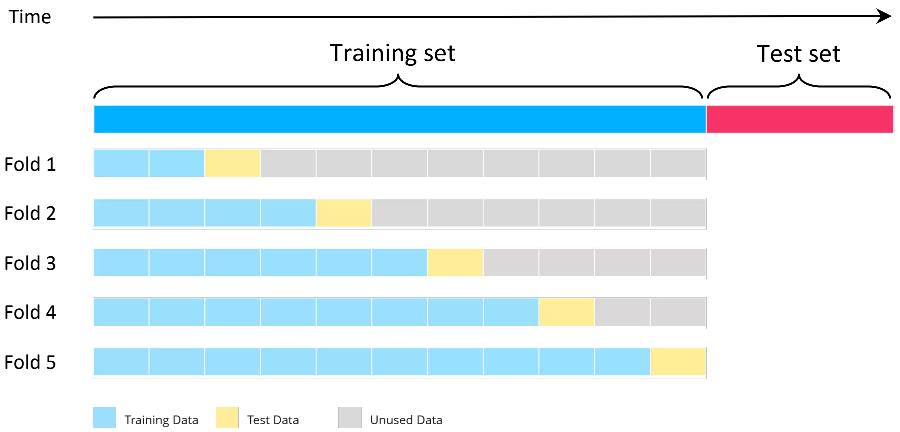
Train / Test split in Time-series data

Never train on the future and test on the past, i.e., **never** use the future to predict the past.



Cross-validation in Time-series data

Never train on the future and test on the past, i.e., **never** use the future to predict the past.



Datasets

Data folder is organized as follows:

- ./stock-historical-data: folder containing historical stock price data.
- ./dividend-history: folder containing historical dividends.
- ./financial-ratio: folder containing financial health of companies.
- ./industry-analysis: folder containing analysis of companies in the same industries.
- companies.csv: list of companies.
- ticker-overview.csv: overview of companies.

Technology stack

A priori, the project is not limited to any tools, libraries and programming languages. Here follows some examples:

Main requirements:

- Python (for programming language).
- Tensorflow (for deep learning and model serving).
- Scikit-learn (for machine learning and data analysis).

Extra credit:

- SQL/MongoDB (for database).
- Airflow (for task orchestration).
- Airbyte (as DB connector).
- dbt (for data transformation).
- Superset (for dashboard).

Final project submission

The structure of submission folder should be organized as follows:

./<StudentID>-project-notebook.ipynb: Jupyter notebook containing source code.
./<StudentID>-project-report.pdf: project report.

The submission folder is named DL4AI-<StudentID>-project (e.g., DL4AI-2012345-project) and then compressed with the same name.

Frequently Asked Questions (FAQs)

Please consult the project description document for detailed answers to the following questions:

1. *What modeling approach is required?*
2. *Do we have to use all the provided data?*
3. *Can we download and use additional external data, such as social trend or sentiment data?*
4. *How should the data be split?*
5. *Can cross validation be used?*
6. *How should hyperparameters and architecture be selected?*
7. *Is feature engineering allowed?*
8. *How should performance be evaluated?*
9. *Is there a required accuracy threshold?*
10. *Do we need to predict the entire market?*
11. *What matters most in grading?*
12. *How detailed should the report be?*
13. *Is extra credit mandatory?*
14. *Can we work in groups?*
15. *Can I also create a demo?*

Deadline

Please visit Canvas for details.

The scope of this project makes it suitable for inclusion in your résumé and job applications.

Thank you